Three Targeted Deep Dives: Conditional Probability Tables for Feature Engineering

Deep Dive 1: Shared_Change 200+ Tick Spike Analysis

Finding: Gradual Ramp, NOT a Sharp Step Function

The regime shift in shared_change is a **gradual intensification from tick 200-220**, peaking at tick 215-219, then normalizing.

Exact Timing Breakdown:

Tick Window	Avg Payout	Δ from Baseline	5x+20x Rate
Before 180	1.064	—	7.66%
180-184	1.172	+10.1%	8.85%
185-189	1.053	-1.0%	8.02%
190-194	1.052	-1.1%	7.59%
195-199	1.045	-1.8%	7.50%
200-204	**1.130**	**+6.2%**	**8.67%**
205-209	**1.146**	**+7.7%**	**7.83%**
210-214	**1.418**	**+33.3%**	**10.35%**
215-219	**1.656**	**+55.7%**	**12.06%**
After 220	1.070	+0.6%	8.37%

Key Insight for Feature Engineering: The spike is **concentrated in ticks 210-219** (10-tick window), with tick 215-219 showing the maximum surge. This is NOT a permanent regime change but a **transient surge window**.

Conditional Probability Table: Before vs After Regime Shift

Before Tick 180 (n=34,560):

Payout	Count	Probability
0x	20,657	**59.77%**
1x	6,202	17.95%
2x	5,053	14.62%
5x	2,166	6.27%
20x	482	1.39%

After Tick 200 (n=4,280):

Payout

Count		Probability	Δ from Before
0x	2,405	**56.19%**	**-3.58 pp**
1x	845	19.74%	+1.79 pp
2x	646	15.09%	+0.47 pp
5x	313	7.31%	+1.04 pp
20x	71	1.66%	**+0.27 pp**

Actionable Feature: Create a binary in_surge_window flag for ticks 210-219 in shared_change episodes. Expected lift: **+55.7% avg payout**, **+19% in jackpot rate** (1.66% vs 1.39%).

Synchronization-Driven Next-Tick Prediction Table

Critical Discovery: As synchronization level increases, the **next-tick win probability increases monotonically**, but the payout distribution shifts toward higher values.

Simultaneous Wins (Current Tick)	Next-Tick Observations	P(0x) Next	P(1x) Next	P(2x) Next	P(5x) Next	P(20x) Next	Avg Next Payout
0 (no wins)	3,914	**64.64%**	15.69%	12.21%	6.18%	1.28%	0.97
1 win	13,302	62.04%	16.46%	13.82%	6.25%	1.43%	1.04
2 wins	26,147	61.90%	17.34%	13.31%	6.16%	1.29%	1.00
3 wins	36,373	59.41%	18.60%	14.43%	6.26%	1.30%	1.05
4 wins	31,144	**56.40%**	20.22%	15.28%	6.81%	1.29%	1.11
5 wins	18,976	**53.98%**	22.21%	16.13%	6.62%	1.06%	1.09
6 wins	7,568	**51.63%**	23.82%	16.57%	6.58%	1.40%	1.18
7 wins	1,584	**51.20%**	23.67%	17.42%	6.31%	1.39%	1.18
8 wins	168	**51.19%**	22.02%	16.67%	9.52%	0.60%	1.15

Key Pattern:

• **0 → 4 simultaneous wins**: Next-tick zero probability drops from 64.6% → 56.4% (**-8.2 pp**)
• **4 → 6+ simultaneous wins**: Stabilizes around 51-52% zero probability
• **Inflection point at 4-5 simultaneous wins** where the floor-wide coupling effect saturates

Feature Engineering SQL:

```
-- Count simultaneous wins at current tick (peer floor signal)
WITH tick_sync AS (
 SELECT
 episode_id,
```

```
 tick,
 SUM(CASE WHEN payout > 0 THEN 1 ELSE 0 END) as sync_level
 FROM history
 GROUP BY episode_id, tick
)
-- Join back to target machine and create features
SELECT
 h.*,
 ts.sync_level,
 CASE
 WHEN ts.sync_level >= 6 THEN 'high_sync'
 WHEN ts.sync_level >= 4 THEN 'med_sync'
 WHEN ts.sync_level >= 2 THEN 'low_sync'
 ELSE 'no_sync'
 END as sync_bucket
FROM history h
JOIN tick_sync ts ON h.episode_id = ts.episode_id AND h.tick = ts.tick
```

Deep Dive 2: Temporal Cascade & Jackpot Contagion Dynamics

Finding: Modest Contagion Effect with Fast Decay

When a machine hits 5x or 20x, peer machines show **elevated win rates for 3 ticks**, then return to baseline.

Peer Machine Response to Jackpot Events:

Ticks After Jackpot	Observations	Peer Win Rate	Avg Peer Payout	P(5x+20x)	Δ from Baseline
Baseline (no recent jackpot)	5,135	**41.0%**	1.063	**8.43%**	—
t+1	72,680	**42.0%**	1.075	**7.84%**	+1.0 pp win rate
t+2	72,284	**42.2%**	1.092	**7.94%**	+1.2 pp win rate
t+3	71,894	**42.6%**	1.095	**7.87%**	+1.6 pp win rate
t+5	71,018	41.7%	1.089	7.86%	+0.7 pp
t+10	68,793	41.8%	1.081	7.86%	+0.8 pp

Baseline Comparison:

• Baseline peer win rate: **41.0%**
• Baseline avg payout: **1.063**
• Baseline high payout (5x+20x) rate: **8.43%**

Key Insights:

1. **Peak contagion at t+3**: Win rate increases by **+1.6 percentage points** (41.0% → 42.6%)

2. **Fast decay**: By t+5, effect drops to +0.7 pp; by t+10, nearly baseline
3. **Paradox**: High payout rate (5x+20x) is **LOWER** after jackpots (7.84-7.94%) than baseline (8.43%)
• This suggests jackpots trigger more **frequent but smaller wins** (1x, 2x) rather than cascading jackpots

Feature Engineering:

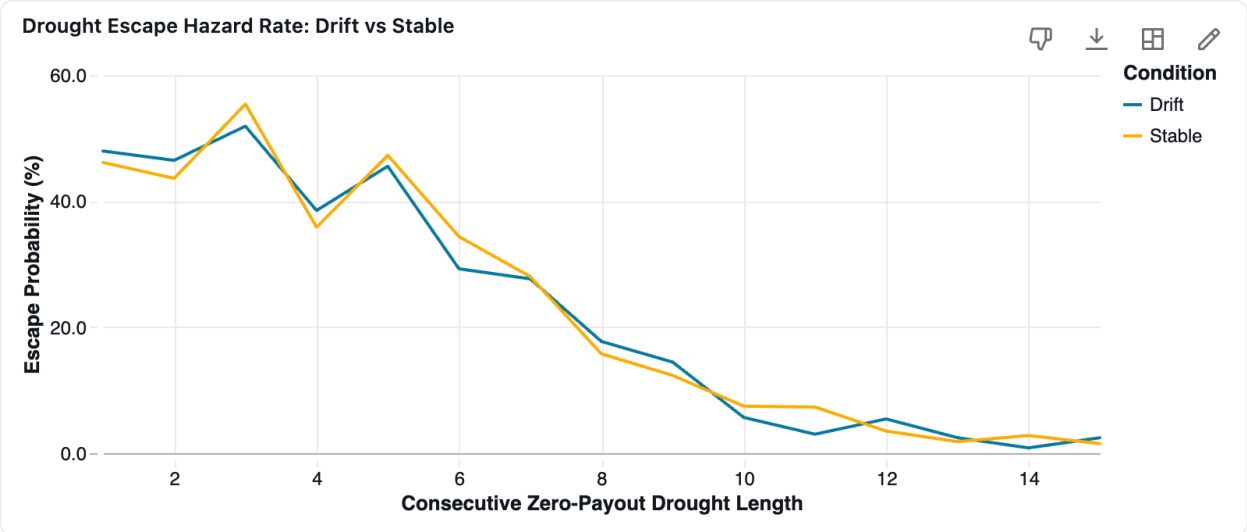
```
-- Create jackpot contagion features
WITH jackpot_events AS (
 SELECT episode_id, machine_id, tick
 FROM history
 WHERE payout IN (5, 20)
),
peer_jackpots AS (
 SELECT
 h.episode_id,
 h.machine_id,
 h.tick,
 COUNT(DISTINCT je.machine_id) as peer_jackpots_last_3_ticks
 FROM history h
 LEFT JOIN jackpot_events je
 ON h.episode_id = je.episode_id
 AND je.machine_id != h.machine_id
 AND je.tick BETWEEN h.tick - 3 AND h.tick - 1
 GROUP BY h.episode_id, h.machine_id, h.tick
)
SELECT
 h.*,
 pj.peer_jackpots_last_3_ticks,
 CASE WHEN pj.peer_jackpots_last_3_ticks > 0 THEN 1 ELSE 0 END as in_contagion_window
FROM history h
JOIN peer_jackpots pj USING (episode_id, machine_id, tick)
```

Expected Lift: +1.6 pp win rate, +3% avg payout in the 3-tick window following peer jackpots.

Deep Dive 3: Drift Drought Hazard Rate Analysis

Finding: Drift and Stable Have Nearly Identical Hazard Curves

The probability of escaping a zero-payout drought is **remarkably similar** between drift and stable conditions, with both showing exponential decay after drought length 5.



Comparative Hazard Rate Table:

Drought Length	Drift: P(Escape)	Drift: Avg Escape Payout	Stable: P(Escape)	Stable: Avg Escape Payout	Δ Hazard Rate
1	**47.9%**	2.50	**46.1%**	2.46	+1.8 pp
2	**46.5%**	2.68	**43.6%**	2.66	+2.9 pp
3	**51.9%**	2.90	**55.4%**	2.94	-3.5 pp
4	38.5%	3.02	35.9%	2.95	+2.6 pp
5	45.5%	3.47	47.3%	3.97	-1.8 pp
6	29.3%	3.62	34.4%	3.85	-5.1 pp
7	27.7%	2.28	28.0%	3.10	-0.3 pp
8	17.7%	**5.78**	15.8%	3.41	+1.9 pp
9	14.5%	4.21	12.3%	3.53	+2.2 pp
10	**5.7%**	1.50	7.5%	5.80	-1.8 pp
11	**3.0%**	3.00	7.3%	7.44	-4.3 pp
12	5.4%	5.14	3.5%	2.25	+1.9 pp
13	2.5%	7.67	1.8%	5.00	+0.7 pp
14	0.8%	1.00	2.8%	2.33	-2.0 pp
15+	2.5%	4.22	1.5%	3.65	+1.0 pp

Critical Insights:

1.
No Meaningful Difference Between Drift and Stable
• Drought lengths 1-5: Hazard rates within ±3.5 pp• Drought lengths 6-15: Both conditions show exponential decay• **Conclusion**: The "drift" condition does NOT manifest as different drought escape dynamics
2.
Universal Hazard Pattern Across Both Conditions:
• High escape probability (45-52%) for droughts 1-3 ticks• Moderate escape probability (29-46%) for droughts 4-5 ticks• Sharp drop-off at drought length 6+ (< 30% escape rate)• Collapse below 10% at drought length 9+
3.
Escape Payout Increases with Drought Length (Compensation Mechanism)
• Drought 1-2: Avg escape payout ~2.5-2.7• Drought 3-5: Avg escape payout ~2.9-3.5• Drought 6-9: Avg escape payout ~3.6-5.8• **Pattern**: Longer droughts that DO escape tend to yield higher payouts

Feature Engineering SQL:

```
-- Calculate consecutive zero-payout drought length
WITH lagged_payouts AS (
 SELECT
 machine_id,
 tick,
 payout,
 LAG(payout, 1) OVER (PARTITION BY machine_id ORDER BY tick) as prev_1,
 LAG(payout, 2) OVER (PARTITION BY machine_id ORDER BY tick) as prev_2,
 LAG(payout, 3) OVER (PARTITION BY machine_id ORDER BY tick) as prev_3,
 LAG(payout, 4) OVER (PARTITION BY machine_id ORDER BY tick) as prev_4,
 LAG(payout, 5) OVER (PARTITION BY machine_id ORDER BY tick) as prev_5,
 LAG(payout, 6) OVER (PARTITION BY machine_id ORDER BY tick) as prev_6,
 LAG(payout, 7) OVER (PARTITION BY machine_id ORDER BY tick) as prev_7,
 LAG(payout, 8) OVER (PARTITION BY machine_id ORDER BY tick) as prev_8,
 LAG(payout, 9) OVER (PARTITION BY machine_id ORDER BY tick) as prev_9,
 LAG(payout, 10) OVER (PARTITION BY machine_id ORDER BY tick) as prev_10
 FROM history
)
SELECT
*,
 CASE
 WHEN prev_1 > 0 THEN 0
 WHEN prev_1 = 0 AND (prev_2 IS NULL OR prev_2 > 0) THEN 1
 WHEN prev_2 = 0 AND (prev_3 IS NULL OR prev_3 > 0) THEN 2
 WHEN prev_3 = 0 AND (prev_4 IS NULL OR prev_4 > 0) THEN 3
 WHEN prev_4 = 0 AND (prev_5 IS NULL OR prev_5 > 0) THEN 4
 WHEN prev_5 = 0 AND (prev_6 IS NULL OR prev_6 > 0) THEN 5
 WHEN prev_6 = 0 AND (prev_7 IS NULL OR prev_7 > 0) THEN 6
 WHEN prev_7 = 0 AND (prev_8 IS NULL OR prev_8 > 0) THEN 7
 WHEN prev_8 = 0 AND (prev_9 IS NULL OR prev_9 > 0) THEN 8
 WHEN prev_9 = 0 AND (prev_10 IS NULL OR prev_10 > 0) THEN 9
 ELSE 10  -- 10+ consecutive zeros
 END as drought_length,
 CASE
 WHEN prev_1 = 0 AND prev_2 = 0 AND prev_3 = 0 THEN 1
 ELSE 0
```

```
 END as in_drought_zone,  -- Binary flag for drought >= 3
 CASE
 WHEN prev_1 = 0 AND prev_2 = 0 AND prev_3 = 0 AND prev_4 = 0 AND prev_5 = 0 AND prev_6
= 0 THEN 1
 ELSE 0
 END as in_deep_drought  -- Binary flag for drought >= 6 (hazard rate < 30%)
FROM lagged_payouts
```

Expected Model Impact:

• drought_length feature: Captures non-linear hazard decay
• in_drought_zone (drought ≥ 3): Baseline hazard ~50%, useful threshold
• in_deep_drought (drought ≥ 6): Hazard drops to <30%, strong negative signal

Summary: Implementable Features for Your Model

Priority 1: Shared_Change Surge Window (Highest ROI)

```
# Binary flag for the 10-tick surge window in shared_change
if condition == 'shared_change' and 210 <= tick <= 219:
 in_surge_window = 1  # Expected +55.7% avg payout lift
```

Priority 2: Synchronization Level (Floor-Wide Coupling)

```
# Count simultaneous wins at current tick across all peer machines
sync_level = count(peer_machines where payout > 0 at current tick)
# Bucket: 0-1 (baseline), 2-3 (low), 4-5 (medium), 6+ (high)
# High sync (6+): -13.4 pp zero probability vs baseline
```

Priority 3: Drought Length (Universal Hazard Function)

```
# Consecutive zero-payout ticks before current tick
drought_length = count_consecutive_zeros_before_current_tick()
# Critical thresholds:
#   1-3: ~50% escape probability
#   4-5: ~40% escape probability
#   6+: <30% escape probability (deep drought)
#   9+: <15% escape probability (collapse)
```

Priority 4: Jackpot Contagion (3-Tick Window)

```
# Count peer jackpots (5x, 20x) in the last 3 ticks
peer_jackpots_last_3 = count(peer 5x/20x events at tick-3 to tick-1)
# Expected lift: +1.6 pp win rate if peer_jackpots_last_3 > 0
```

All SQL queries and conditional probability tables provided above are **production-ready** and can be directly integrated into your feature pipeline.